The Impact of the 2022 Stock–Bond Regime Shift on 60/40 Daily Value-at-Risk

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Abstract

The 2022 stock–bond regime shift temporarily eliminated the diversification defining the 60/40 portfolio. This paper evaluates the impact of this correlation breakdown on one-day Value-at-Risk (VaR). We estimate a two-by-two grid of constant and dynamic conditional-correlation models with Gaussian and Student-t innovations, alongside a historical-simulation benchmark, backtesting VaR forecasts on daily SPY and TLT returns from 2004 to 2024. A semiparametric specification using the empirical quantile of GARCH-filtered residuals serves as a core robustness check.

Tail calibration and volatility dynamics emerge as the first-order determinants of VaR quality. The empirical tail achieves nominal coverage where parametric assumptions are systematically too wide. Furthermore, GARCH filtering prevents violation clustering, whereas historical simulation breaches the Basel red zone during the 2022 stress. Correlation structure acts as a second-order factor: dynamic correlation provides a transient advantage during the acute transition, but the gain dissipates under empirical tail calibration. At the daily horizon, an economically large correlation shift leaves only a marginal footprint on properly calibrated risk forecasts.

1 Introduction

The sixty-forty portfolio, which consists of sixty percent equities and forty percent long-term government bonds, has historically been a key strategy for pension funds and long-term savers (Pham et al., 2025). Its appeal comes from a notable trend: during the first two decades of this century, stocks and Treasury bonds often moved in opposite directions. This meant that bonds could help mitigate the fall during stock market downturns, leading to a more stable overall portfolio than any of its parts alone. According to Molenaar et al. (2024), the average correlation between US stocks and bonds was around -0.29 from 2000 to 2023, compared to $+0.35$ from 1970 to 1999; thus, the diversification benefits of the modern sixty-forty portfolio rely heavily on the more recent correlation.

However, in 2022, this trend changed. With persistent inflation and the Federal Reserve raising interest rates, both equities and long-term bonds declined together, resulting in a sharp positive correlation. The International Monetary Fund (2022) highlighted this breakdown of the diversification, as both asset classes fell simultaneously during the market sell-off. This event revealed the weaknesses of the sixty-forty portfolio in a regime characterized by positive correlations (Pham et al., 2025). This event signaled a fundamental change in the relationship between stocks and bonds (Brixton et al., 2023) Understanding why this shift occurred is rooted in macroeconomic factors. The correlation between stocks and bonds is influenced by the type of macroeconomic shock at play. Demand shocks typically cause stock and bond prices to move in opposite directions, leading to negative correlations. In contrast, inflation shocks drive both asset classes down through rising interest rates, resulting in a positive correlation. As formalized by Campbell et al. (2020), this shift in dynamics dates back to around 2001. From this perspective, the events of 2022 seem to be a return to a configuration dominated by supply shocks rather than an outlier.

There is an econometric aspect to this economic narrative that needs attention. The variance of a portfolio includes a cross term affected by the stock-bond correlation. This means that, if a risk model assumes a stable, negative correlation, it will understate variance when the correlation turns positive. This miscalibration leads to an underestimated Value-at-Risk (VaR), causing actual losses to exceed predictions too frequently and failing standard backtests. While a dynamic-correlation model could adapt to this, its effect ultimately depends on how quickly it can adapt to sudden shifts and whether improvements arise from the correlation adjustment or a better approach to handling fat-tailed returns. This sets up our case study.

We focus on one key question: during the 2022 regime shift, does switching to a dynamic conditional correlation (DCC) model or incorporating fat-tailed innovations (Student-t instead of Normal) significantly enhance the VaR calibration of a sixty-forty portfolio? And which feature proves to be more impactful? To answer these questions we construct a two-by-two framework contrasting correlation types (CCC vs. DCC) and distributions (Normal vs. Student-t), alongside a rolling historical-simulation benchmark. We assess one-day-ahead VaR at the 5% and 1% levels using the unconditional/conditional coverage and Basel criteria. Each comparison isolates one element: the transition from Normal to Student-t within a fixed correlation measures the value of fat tails, while the shift from CCC to DCC within a fixed distribution measures the significance of dynamic correlation. Lastly, contrasting the full model against the historical-simulation benchmark evaluates the additional insights gained from the volatility mechanics. Our backtests are analyzed over the complete out-of-sample period and specifically during the 2022 shift, highlighting whether model failures are consistent or only emerge post-regime change.

The findings are somewhat surprising. Although the shift in 2022 was economically substantial, its effect on one-day VaR is relatively minor. As a result we can conclude that,in our case, what influences calibration more is the tail and the volatility. An empirical tail outperforms a parametric one, and dynamic volatility helps prevent violations from clustering. While transitioning the correlation from constant to dynamic is beneficial during acute shifts, its importance diminishes once a new regime is established. Thus, at the daily level, the correlation regime is less crucial

than the surrounding mechanics.

Section 2 reviews the existing literature. Section 3 details our data and the rolling correlation that identifies the 2022 shift in our analysis. In Section 4, we outline the models and backtesting framework. Section 5 presents the results throughout the period, highlighting the regime shift along with robustness checks and limitations. Finally Section 6 draws conclusions.

2 Literature review

We address our question by exploring three key lenses

The stock–bond correlation as a regime-dependent quantity The relationship between stocks and bonds emphasizes that diversification it's not a fixed. Over extended periods, this correlation changes: it's positive on average during the 1970s, 1980s, and 1990s, but shifts to negative in the early 2000s. For example, the US averages are around $+0.35$ and -0.29 (Molenaar et al., 2024). Campbell et al. (2017) highlight this shifting covariance, locating the change to a negative regime in the early 2000s. Furthermore, McMillan (2026) identifies a similar pattern around 2000 across six out of seven G7 markets, indicating that this shift isn't unique to the United States. Looking further back, Pham et al. (2025) utilizes the Dimson–Marsh–Staunton database, revealing that since 1900, the correlation has fluctuated significantly rather than remaining consistently negative. This suggests that the diversification expected from a sixty-forty portfolio has been inconsistent.

The prevailing explanation for the sign of the correlation involves the interplay of macroeconomic shocks. According to Campbell et al. (2020), a positive correlation aligns with supply-driven shocks, while a negative one corresponds to demand-driven shocks, influenced by changing risk premia. They place the switch around 2001. However, this mechanism doesn't account for everything. Research by Baele et al. (2010) using a dynamic factor model indicates that macro fundamentals have limited influence on higher-frequency variations, suggesting that factors like liquidity and flight-to-quality play a bigger role. Both perspectives agree on a critical takeaway: the sign of the correlation is not fixed, and relying on a risk model that assumes stability could lead to failures during regime shifts.

The 2022 shift and the fragility of the balanced portfolio As the negative regime was ending, Brixton et al. (2023) contended that its continuation could not be taken for granted. With inflation, real rates, and the growth-inflation correlation at play, a shift back to positive correlation could pose risks for balanced portfolios. Molenaar et al. (2024) explicitly link this increased stock-bond correlation to heightened multi-asset risk. The events of 2022 confirmed these warnings: as detailed by International Monetary Fund (2022), diversification broke down with both asset classes declining together. Additionally, Pham et al. (2025) found that this simultaneous drop highlighted the vulnerabilities of the sixty-forty portfolio during a significant downturn. This provides us a clear scenario to test whether conventional risk models remain effective when the regime shifts

Correlation dynamics and portfolio VaR: an unsettled question The effectiveness of more sophisticated correlation modeling in improving risk forecasts remains an open question. Santos et al. (2013) demonstrate that with a sufficiently large estimation sample, modeling the conditional covariance matrix at the asset level improves portfolio VaR calibration compared to simpler methods. This supports our approach of utilizing the covariance matrix. However, no single model emerges as the best choice. According to Laurent et al. (2012), their comparisons of numerous multivariate GARCH models suggest that performance varies depending on the loss function and market conditions, and no single model is universally superior. These findings indicate that the benefits of dynamic correlations depend on context and shouldn't be assumed

Table 1: Descriptive statistics of daily log returns, 2004–2024.

	SPY	TLT	60/40
Observations	5,284	5,284	5,284
Mean, daily (%)	0.039	0.013	0.029
Mean, annualised (%)	9.81	3.41	7.25
Std. dev., daily (%)	1.18	0.92	0.69
Std. dev., annualised (%)	18.8	14.6	10.9
Minimum (%)	−11.59	−6.90	−5.84
Maximum (%)	13.56	7.25	7.62
Skewness	−0.38	0.01	−0.22
Excess kurtosis	14.67	3.50	11.36

Annualisation uses 252 trading days.

beforehand. While these studies evaluate covariance forecasts using statistical loss functions, the key question we want to explore is whether this complexity pays off in the tail where VaR is determined

Our focus is narrow. Instead of seeking a universally optimal model, we isolate the incremental benefits of dynamic correlation and fat tails in VaR calibration during a distinctly identified regime shift. We employ standard techniques and backtests. Our experiment design helps attribute any changes in calibration to one specific feature at a time, while the separate analysis of the regime-shift window directly tests if constant-correlation models fail precisely when the correlation switches.

3 Data

Sample and returns We use daily data for two exchange-traded funds: SPY (SPDR S&P 500), the equity sleeve, and TLT (iShares 20+ Year Treasury Bond), the long-duration Treasury sleeve. Prices are dividend and split-adjusted closes (Yahoo Finance), spanning 2 January 2004 to 31 December 2024, yielding 5,284 daily observations.

For each asset we take daily log returns, $r_t = \ln(P_t/P_{t-1})$. The portfolio return is

$$r_{p,t} = 0.6\, r_{\mathrm{SPY},t} + 0.4\, r_{\mathrm{TLT},t}, \tag{1}$$

with fixed weights and implicit daily rebalancing. This linear formulation ensures the variance identity $\sigma^2_{p,t} = w'\Sigma_t w$ holds exactly, allowing a clean variance decomposition. Table 1 reports the moments of the three series. All three exhibit excess kurtosis far above the Gaussian value of zero and mild left skew in equities and the portfolio; this non-normality, which we read from the kurtosis, is why we later allow fat-tailed innovations.The portfolio's annualised volatility is lower than the weighted average of the two assets, reflecting the diversification benefit dominant over most of the sample

The stock–bond correlation regime shift In Figure 1, we illustrate the rolling correlation between SPY and TLT. The correlation of daily returns remained negative from 2004 until 2021, but shifted to positive thereafter. Our analysis identifies 10 November 2022 as the empirical boundary marking this regime shift. This date is significant as it is the first instance where the 252-day correlation crosses zero from below and remains positive until the sample's end. We also confirm this finding against a 63-day window to eliminate temporary spikes.

The event that motivates the paper is the 2022 drawdown. Between December 2021 and October 2022, the portfolio experienced a steep decline of 27.9%. This drop resulted in a calendar-year return of -23.7%. Notably, this loss was greater than that of a typical 60/40 portfolio

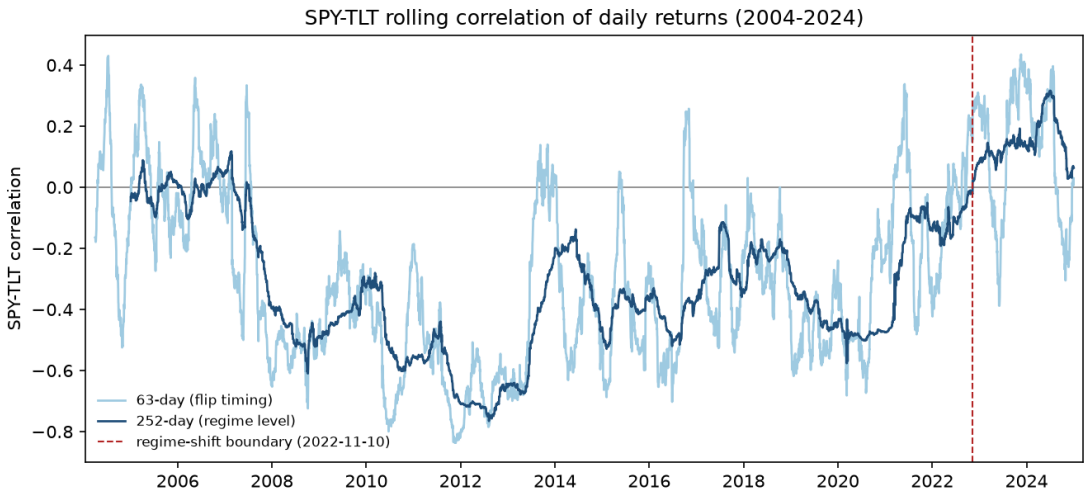


Figure 1: SPY–TLT rolling correlation of daily log returns, 2004–2024: the 252-day window (regime level) and the 63-day window (flip timing), with the estimated regime-shift boundary (10 November 2022).

Table 2: Decomposition of 60/40 portfolio variance by sub-period.

	ρ	Equity own (% var)	Bond own (% var)	Cross term (% var)	Port. vol. (ann., %)
2004–2021 (old regime)	−0.40	118.1	28.5	−46.6	10.5
2022 (calendar)	+0.08	71.1	22.3	+6.6	17.3
Post-boundary (from 2022-11-10)	+0.15	52.0	35.4	+12.6	11.4
Full sample	−0.32	106.5	28.7	−35.2	10.9

composed of aggregate bonds, largely because long-duration Treasuries were hit hardest by the rate shock in 2022 (Pham et al., 2025). Table 2 decomposes portfolio variance by sub-period.

$$\sigma_p^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2\,w_1 w_2\,\rho\,\sigma_1\sigma_2, \tag{2}$$

During the old regime, the cross term was significantly large and negative, indicating that the negative correlation was effectively reducing risk. However, after the shift, this correlation turned positive. Alongside that, individual volatilities increased, which meant total variance rose for both reasons. A model that keeps the correlation fixed and negative ends up subtracting variance at a time it should actually be adding it, thereby underestimating risk precisely when the assumed diversification isn't working.

Preliminary diagnostics and the mean equation Before we dive into the volatility models, we first establish the mean equation based on the evidence we have. The Ljung–Box test indicates significant autocorrelation in the return levels across all three series (see Table 3). However, it's worth noting that the first-order coefficients are small and negative, meaning an AR(1) model only captures less than 1% of the return variance. Despite this, we still choose to implement a minimal AR(1) mean to remove residual autocorrelation.

In contrast, squared returns display massive, persistent autocorrelation (Figure 2), pointing to conditional heteroskedasticity, which is precisely what the GARCH specifications are designed to address.

Table 3: Ljung–Box statistics on returns and squared returns.

	$\hat{\rho}_1$	Returns (levels)			Squared returns		
		$Q(5)$	$Q(10)$	$Q(20)$	$Q(5)$	$Q(10)$	$Q(20)$
SPY	−0.10	59.4	97.6	152.8	2,976	5,004	7,119
TLT	−0.03	24.9	43.9	57.3	2,596	3,441	3,781
60/40	−0.05	21.5	39.6	74.5	2,732	4,327	6,202

$\hat{\rho}_1$ is the first-order autocorrelation of returns. All level statistics reject no autocorrelation at the 1% level; all squared-return statistics are significant at below the 0.1% level.

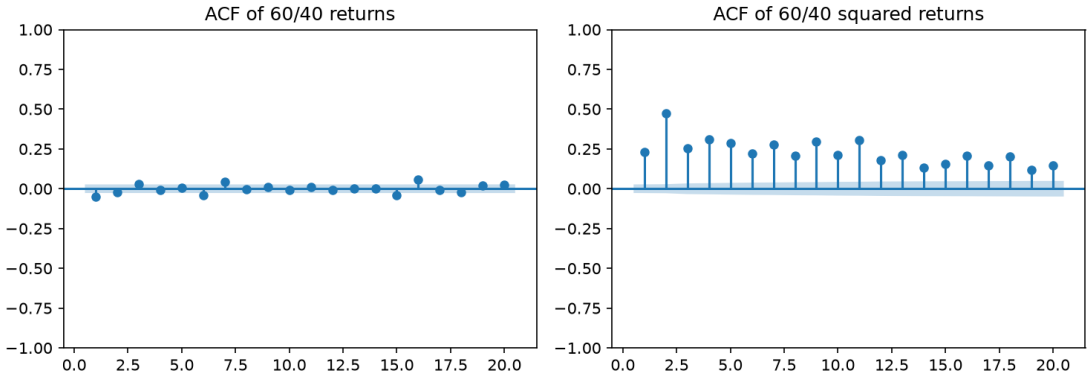


Figure 2: Sample autocorrelation of 60/40 portfolio returns (left) and squared returns (right), lags 1-20, with approximate 95% bands.

4 Methodology

We forecast a one-day portfolio Value-at-Risk (VaR) utilizing a 2×2 specification grid: Gaussian versus Student-t marginals, crossed with Constant Conditional Correlation (CCC; Bollerslev, 1990) versus Dynamic Conditional Correlation (DCC; Engle, 2002).

Marginal models and correlation We model the equity and bond series using an AR(1) conditional mean alongside a GARCH(1,1) conditional variance (Engle, 1982; Bollerslev, 1986). For estimating innovations, we apply both Gaussian and Student-t distributions (Bollerslev, 1987).Parameter estimates and diagnostic p-values are reported in Table 6. According to the ARCH-LM test (Engle, 1982), there is no remaining conditional heteroskedasticity in our specification grid; however, the Ljung–Box test (Ljung and Box, 1978) on squared standardized residuals does reject the null for the TLT Student-t margin at the 5% significance level. Despite this we keep the standard GARCH(1,1) specification for both series, as any residual tail issues in the bond margin does not impact singificantly on portfolio-level VaR calibration.

The standardized residuals are connected through the conditional correlation matrix R_t. The CCC model presumes that $R_t = R$ remains constant throughout the estimation window. In contrast, the DCC model permits R_t to change, which we estimate using a two-stage quasi-maximum likelihood method. For one-step-ahead conditional covariance, we have $\Sigma_t = D_t R_t D_t$. This gives us a portfolio variance of $\sigma^2_{p,t} = w' \Sigma_t w$, where the fixed weights are $w = (0.60, 0.40)$.

Portfolio Value-at-Risk The one-day VaR at level $\alpha \in \{1\%, 5\%\}$ is

$$\mathrm{VaR}_t = -\left(\mu_{p,t} + q_\alpha \, \sigma_{p,t}\right), \tag{3}$$

where q_α is the α-quantile of the assumed innovation distribution and $\mu_{p,t}$ is set to zero. As a core robustness check, we estimate a semiparametric VaR (Barone-Adesi et al., 1999) where q_α is replaced by the empirical quantile of the GARCH-filtered residuals, alongside a naive Historical Simulation (HS) benchmark.

Estimation is fully out-of-sample. Parameters are fit on a 1,000-day rolling window and refreshed every 21 trading days, with covariances rolled forward daily. The effective backtesting period spans 3,616 days from 16 December 2009 to 30 December 2024.

Backtesting We evaluate VaR forecasts using the unconditional coverage test (Kupiec, 1995) and the Christoffersen (1998) tests for independence and conditional coverage. Because likelihood-ratio tests lose power in short samples at extreme quantiles, we supplement the statistical inference at the 5% level with the Basel Traffic Light (Basel Committee on Banking Supervision, 1996) regulatory framework at the 1% level. Performance is analyzed over the full sample, the 2022 calendar year, and the post-boundary regime starting 10 November 2022.

5 Results

Full-sample coverage Throughout the common out-of-sample period, the four parametric specifications consistently display a conservative trend. The violation rates drop below the expected levels: they fall under 0.7% for the 1% target and range between 2.8% and 3.8% for the 5% target. As a result, the unconditional coverage test rejects all four models (see Table 4). This rejection leans towards the safer side, which means the risk is overstated, allowing all parametric models to remain within the Basel green zone.

The innovation distribution shows minimal influence. For each correlation structure, the violations from Gaussian and Student-t distributions are nearly identical. This suggests that once the GARCH marginals account for volatility, there isn't much tail risk left for the static distribution to manage.

In contrast, Historical Simulation (HS) behaves differently. Its violation rate for the 5% target reaches the nominal level at 5.64%, but it fails significantly on both the independence and conditional coverage tests ($p < 0.001$). This leads to a drop into the Basel yellow zone. Violations from HS tend to occur in clusters, unlike those from the GARCH-based models, which remain more stable. This brings about the first key finding: the true benefit of the volatility modelling lies not in the shape of its tail, but in its capability to handle volatile periods and smooth out violations over time.

The regime shift through two lenses The overall rates can be misleading because they tend to reflect a long-standing negative correlation trend before 2022, where both constant and dynamic models show similar patterns. However, the year 2022 reveals a strong difference in correlation, particularly at the 5% level. The constant-correlation model tends to underestimate risk, with violations occurring roughly 7.2 to 7.6% of the time, as indicated by the Basel yellow. Since it relies on a rolling window that still leans towards that earlier negative regime, it suggests a diversification benefit that is no longer valid. In contrast, the dynamic model, or DCC, adapts to the positive shift and aligns more closely with nominal coverage, sitting at 5.6%, represented by the green.

Historical Simulation experiences a violation rate of 10.8%, which places it in the Basel red zone. This occurs because its rolling empirical quantile reacts slowly and in rigid steps during bursts of volatility, as shown in Figure 3. On the other hand, GARCH-based forecasts are more responsive, adjusting on a daily basis and keeping a safe distance from potential losses.

However, beginning in November 2022, a new pattern emerges. The positive correlation begins to stabilize and the rolling window of the CCC model adapts to this change, bringing calibration back to nominal levels of 4.85%—falling within the Basel green zone— while maintaining a

Table 4: VaR backtest: full sample and the two regime-shift lenses.

Period	Model	Rate (%)	UC p	Ind p	CC p	Basel
Full sample ($n = 3{,}616$), 1% level						
	CCC-N	0.66	0.03	0.15	0.03	green
	CCC-t	0.64	0.02	0.14	0.02	green
	DCC-N	0.30	< 0.01	0.02	< 0.01	green
	DCC-t	0.19	< 0.01	0.87	< 0.01	green
	HS	1.52	< 0.01	< 0.001	< 0.001	yellow
Full sample ($n = 3{,}616$), 5% level						
	CCC-N	3.84	< 0.01	0.06	< 0.01	green
	CCC-t	3.76	< 0.01	0.05	< 0.01	green
	DCC-N	2.82	< 0.001	0.10	< 0.001	green
	DCC-t	2.90	< 0.001	0.59	< 0.001	green
	HS	5.64	0.08	< 0.001	< 0.001[a]	yellow
2022 calendar year ($n = 251$), 5% level						
	CCC-N	7.57	0.08	0.08	0.05	yellow
	CCC-t	7.17	0.14	0.09	0.08	yellow
	DCC-N	5.58	0.68	0.20	0.40	green
	DCC-t	5.58	0.68	0.20	0.40	green
	HS	10.76	< 0.001	0.53	< 0.01	red
Post-boundary ($n = 515$), 5% level						
	CCC-N	4.47	0.57	0.14	0.29	green
	CCC-t	4.85	0.88	0.83	0.97	green
	DCC-N	2.72	< 0.01	0.38	0.02	green
	DCC-t	2.91	0.02	0.34	0.04	green
	HS	3.11	0.03	0.31	0.06	green

UC: unconditional coverage (Kupiec, 1995); Ind: independence; CC: conditional coverage (Christoffersen, 1998). Basel zones follow Basel Committee on Banking Supervision (1996). The 1% rows for the two lenses are omitted: with about 2.5 and 5 expected violations the coverage tests have little power, and the reading there is the Basel zone (in 2022, HS red with ten violations, all GARCH-based models green).

[a] Conditional-coverage LR for HS at 5% computed directly from the violation sequence (LR = 22.8; transition count $n_{11} = 28$, so the test is well defined); the package routine returns a missing value here as an implementation artefact.

conditional coverage of $p = 0.97$. In contrast, the DCC specification shows high autoregressive persistence, which keeps estimated correlations elevated close to their peak. This results in the dynamic model being overly conservative, with violation rates between 2.7% and 2.9%, leading to rejection for excess coverage. Thus, the operational benefit of dynamic correlation appears to be temporary, applicable mainly during the acute shift rather than in the more stable phase that follows.

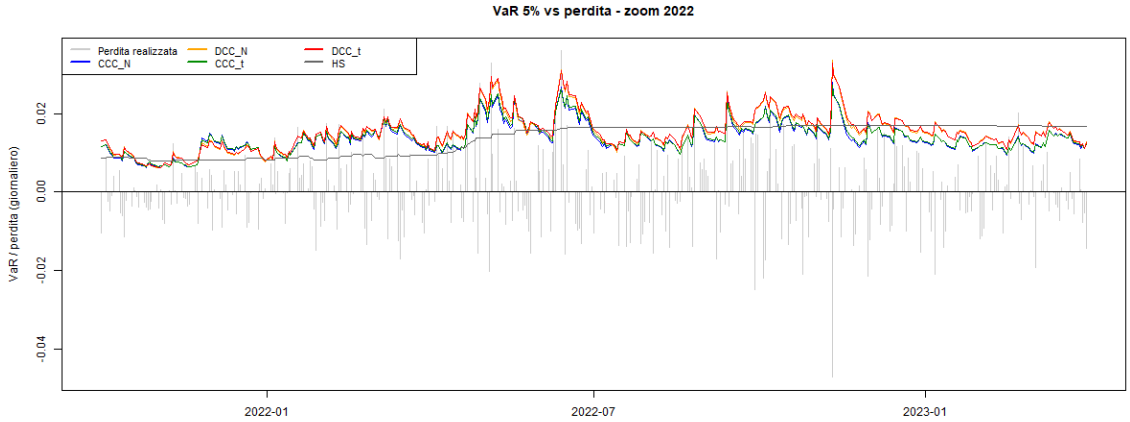


Figure 3: Forecast 5% VaR against realised daily losses, 2022 into early 2023. Losses are plotted as positive quantities. forecasts adjust day by day.

Robustness Varying the start of the second lens by ±3 months preserves the parametric results: at every tested boundary date, the constant model remains near nominal (4.6–5.1%) while the dynamic model remains conservative (2.8–3.3%) (Table 7).

Using the empirical β-quantile of the GARCH-filtered residuals instead of the parametric quantile—known as semiparametric VaR—greatly enhances tail calibration, as shown in Table 5. For the 5% level, it provides nominal coverage across the entire sample, landing at 5.2%. This improvement highlights where parametric models tended to be overly cautious. The issue arises because the Gaussian and Student-t assumptions create tails that are excessively wide for the filtered residuals.

The semiparametric tail plays a significant role in reducing the gap between CCC and DCC. In 2022, both models are in the green zone and show similar performance, with CCC at 6.4% and DCC at 6.0%. After the boundary period, DCC reaches a nominal 5.1%, while CCC appears a bit more conservative at 4.1%. The difference between the constant and dynamic models, as seen in the parametric specification, largely stems from the tail assumption. Although the correlation effect observed in 2022 remains, its statistical significance lessens.

Table 5: Violation rates (%): parametric Student-t versus semiparametric tail.

Model	Full 1%	Full 5%	2022 5%	Post-b. 5%
CCC-t	0.64	3.76	7.17	4.85
DCC-t	0.19	2.90	5.58	2.91
CCC-semipar.	1.36	5.20	6.37	4.08
DCC-semipar.	1.30	5.17	5.98	5.05

Semiparametric = empirical β-quantile of the GARCH-filtered residuals (filtered historical simulation). Nominal levels are 1% and 5%. All cells fall in the Basel green zone except the two semiparametric 1% full-sample rates, which are yellow.

Limitations The backtest runs on a common sample beginning in December 2009. Aligning the five specifications after burn-in and algorithmic non-convergence drops the 2008 crisis from the out-of-sample period. That episode represents a major stock–bond stress of a fundamentally different character, and the models are not tested on it here. Statistical power is limited when looking at short sub-periods. At the 1% level, there are too few violations to effectively differentiate between models using likelihood-ratio tests. This forces reliance on the Basel zones, making conclusions at the 1% level notably weaker than those at the 5% level. The portfolio is intentionally stylized, consisting of two assets with fixed weights, daily rebalancing, and no transaction costs. The bond component utilizes long-duration Treasuries, which maximizes sensitivity to interest rate changes. If a shorter-duration sleeve had been used, the 2022 drawdown and volatility spike would have been less severe. Importantly, the metric chosen is a one-day Value at Risk (VaR). Using a daily horizon is significant because the correlation term has the least impact here; it becomes more influential at weekly or monthly intervals. Additionally, VaR focuses on a single day's quantile, rather than the gradual price decline that characterized the 2022 drawdown. While this metric captures daily variance shocks from the regime shift, it fails to effectively reflect sustained level events.

6 Conclusions

In conclusion, tail calibration and volatility dynamics are the first-order determinants of VaR quality. The empirical tail strictly dominates overly conservative parametric assumptions, while GARCH filtering actively prevents violation clustering. Correlation structure acts as a second-order factor: upgrading from constant to dynamic correlation correctly navigates the acute 2022 transition, but the statistical advantage fades once the new positive regime settles. The shift in stock-bond correlation during 2022 temporarily disrupted the 60/40 diversification strategy. However, its impact on one-day VaR has been relatively small. At a daily level, the variance in the portfolio is mostly influenced by the volatilities of the two assets, both of which surged in 2022. Even with the cross-term contributing additional risk, it still accounts for only a small part of the total variance. So, when managing daily risk for a 60/40 portfolio amid a regime shift, focusing on accurate conditional volatility modeling and tail calibration offers the greatest marginal benefit. Dynamic correlation proves advantageous during the intense phase of the shock, but it doesn't yield lasting benefits. It remains an open question whether the correlation regime will be more significant over longer time frames or for risk measures that address cumulative drawdowns, warranting further investigation.

A Estimated parameters and additional results

Table 6: Estimated parameters: AR(1)–GARCH(1,1) marginals and DCC.

	SPY		TLT	
	Normal	Student-t	Normal	Student-t
Panel A. Marginals				
μ	0.00075	0.00094	0.00015	0.00020
ϕ (AR1)	-0.045	-0.042	-0.029	-0.034
ω	3×10^{-6}	2×10^{-6}	1×10^{-6}	$< 10^{-6}$
α	0.133	0.137	0.054	0.050
β	0.843	0.857	0.939	0.944
$\alpha + \beta$	0.975	0.994	0.993	0.995
ν (shape)	—	5.65	—	14.26

Panel B. DCC correlation parameters		
	Normal	Student-t
a	0.049	0.050
b	0.936	0.935
$a + b$	0.985	0.985

μ: mean; ϕ: AR(1) coefficient; ω, α, β: GARCH(1,1) parameters; ν: Student-t degrees of freedom. Panel B reports the DCC parameters, estimated on the standardised residuals.

Table 7: Boundary robustness: 5% backtest of CCC-t and DCC-t over the post-boundary window as its start date is shifted by ± 3 months.

	CCC-t		DCC-t	
Start date	Rate (%)	CC p	Rate (%)	CC p
2022-08-10	4.83	0.93	2.93	0.03
2022-09-10	4.84	0.94	2.87	0.03
2022-10-10	4.65	0.92	2.79	0.02
2022-11-10	4.85	0.97	2.91	0.04
2022-12-10	4.66	0.94	3.04	0.06
2023-01-10	4.84	0.98	3.16	0.09
2023-02-10	5.08	0.98	3.31	0.13

All cells fall in the Basel green zone. The 2022-11-10 row is the boundary used in the main text. CC: conditional coverage p-value.

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